



Minwoo Kim

Date of birth: 19/06/1995 | **Nationality:** South Korean (Republic of Korea) | **Phone number:** (+82) 1052979385 (Mobile) | **Email address:** mwoo.kim@samsung.com | **Website:** <https://fair.kaist.ac.kr/> | **LinkedIn:** <https://www.linkedin.com/in/minwoo-kim-4a2468208/> | **github:** <https://github.com/mw9385> | **Address:** South Korea (Work)

ABOUT ME

Minwoo Kim is a dedicated researcher with extensive experiences in autonomous mobile robots, unmanned vehicles (e.g., UAVs or UGVs), precise robot manipulation for humanoids research working in the Samsung Research in South Korea. His special expertise lies in the area of vision-based navigation, reinforcement learning and inverse reinforcement learning problems, UAVs/UGVs technology applications such as rescue, delivery, and inspection. He has strong academic publication records including journals, and conference papers in relation to autonomous mobile robots.

While his core expertise is rooted in vision-based navigation and reinforcement learning, he is currently actively engaged in pioneering research on Vision-Language-Action (VLA) models for real robots. By integrating his background in autonomous systems with multimodal foundation models, he is driving new advancements in robotic intelligence. His contributions are well-recognized through a strong record of publications in top-tier journals and conferences.

EDUCATION AND TRAINING

03/2019 – 28/02/2025 Ulsan, South Korea
COMBINED MASTER'S AND PH.D Ulsan National Insitute of Science and Technology

Thesis Learning-Based Visual Navigation for Drones: Generalization, Sample-Efficiency and Sim-to-Real Gap

03/2014 – 02/2019 Ulsan, South Korea
MECHANICAL AND AEROSPACE ENGINEERING (BACHERLOR'S DEGREE) Ulsan National Insitute of Science and Technology

Field of study Mechanical Engineering, Business Management | **Final grade** Magna cumme laude

WORK EXPERIENCE

STAFF ENGINEER – SAMSUNG RESEARCH – 01/04/2025 – Current – SEOUL, SOUTH KOREA

- Robot Intelligence Team
- Vision-Language-Action (VLA) model for precise manipulation
 - VLA finetuning using Reinforcement Learning

ENGLISH LECTURER – SAMSUNG WALFARE – 09/2016 – 02/2017

As a type of educational volunteer, provides educational opportunity to underprivileged middle school students. It is funded by SAMSUNG Electronics. They also provided us a various learning experience especially how to treat young students and teaching methods.

PROJECT MANAGER – ENACTUS KOREA – 03/2014 – 03/2017

Link : <https://platum.kr/archives/36219>

INTERNSHIPS – ULSAN NATIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY – 01/2018 – 01/2019

HONOURS AND AWARDS

12/2021
Best Intelligent Paper Award – International Conference on Robot Intelligence Technology and Applications (RiTA)

Research topic: Uncertainty-Aware Time-To-Collision Prediction Using Convolutional Long-Short-Term Memory Networks

10/2019
Grand Prize at the 2019 AI Competition for Space Weather Disaster Prediction – National Radio Research Agency within the Ministry of Science and ICT (MSIT)

Link <https://news.unist.ac.kr/unist-graduate-students-honored-by-national-radio-research-agency-for-research-excellence/>

11/2021
Won a gold prize in Ulsan-Challenge Festival – Ulsan National Institute of Science and Technology

Take on the role of a project manager. Team ASL (under supervision of Prof. Oh and Dr. Lim in PABLO AIR) won a gold prize for their BTS (brain-to-society) project on situation awareness using vision and machine learning for safe drone/UAM operation.

Link <https://news.unist.ac.kr/kor/20211203-2/>

● **LANGUAGE SKILLS**

Mother tongue(s): **KOREAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **SKILLS**

Python | Pytorch | Tensorflow | Machine Learning | Deep Learning | Computer Vision | Deep Reinforcement Learning | Robot Operating System

● **PUBLICATIONS**

2026
[Vision-Based Autonomous Drone Landing on Moving Platforms With Uncertain Motion via Deep Reinforcement Learning](#)

Authors: Woojae Shin, Minwoo Kim, Taewook Park, Geunsik Bae, Seunghwan Kim, Hyondong Oh | **Journal Name:** IEEE Robotics Automation Letters

2025
[RAPID: Robust and Agile Planner Using Inverse Reinforcement Learning for Vision-Based Drone Navigation](#)

Authors: Minwoo Kim, Geunsik Bae, Jinwoo Lee, Woojae Shin, Changseung Kim, Myongyol Choi, Heejung Shin, Hyongdong Oh | **Journal Name:** Robotics Science and Systems

2024
[Collective Navigation Through a Narrow Gap for a Swarm of UAVs Using Curriculum-Based Deep Reinforcement Learning](#)

Authors: Myong-Yol Choi, Woojae Shin, Minwoo Kim*, Hwi-Sung Park, Youngbin You, Min Lee, and Hyondong Oh | **Journal Name:** Korea Robotics Society

2022
[Exploration in deep reinforcement learning: A survey](#)

3 year citations: 753

Authors: Pawel Ladosz, Lilian Weng, Minwoo Kim and Hyondong Oh | **Journal Name:** Information Fusion

2022
[Towards monocular vision-based autonomous flight through deep reinforcement learning](#)

Authors: Minwoo Kim, Jongyun Kim, Minjae Jung and Hyondong Oh | **Journal Name:** Expert Systems with Applications

2022
[Monocular vision-based time-to-collision estimation for small drones by domain adaptation of simulated images](#)

Authors: Minwoo Kim, Pawel Ladosz, Hyondong Oh | **Journal Name:** Expert Systems with Applications

● CONFERENCES AND SEMINARS

07/2023 Robotics: Science and Systems, workshop.

Learning From the Best: Inverse Imitation Learning for Agile Drone Flight

11/2022 In Autumn Conference of Korean Society for Aeronautical and Space Sciences, Jeju, South Korea

Vision-Based Highspeed Collision-Free Trajectory Generation Using Sample Efficient Inverse Reinforcement Learning.

05/2022 Asian Control Conference, Jeju, South Korea (presentation-only option)

Boustrophedon coverage path planning using deep reinforcement learning based traveling salesman problem

11/2020 In Autumn Conference of Korean Society for Aeronautical and Space Sciences, Jeju, South Korea

Learning-Based Algorithm Framework for Drone Trail Following with Obstacle Avoidance

11/2020 In Autumn Conference of Korean Society for Aeronautical and Space Sciences, Jeju

Vision-Based Time-to-Collision Prediction for Small Drones by Domain Adaptation of Simulated Images

03/2020 The 15th Korea Robotics Society Annual Conference (Online Participation)

Real time human detection on construction site: deep learning framework on 3D point cloud data for human detection

11/2019 In Autumn Conference of Korean Society for Aeronautical and Space Sciences, Jeju, South Korea

Collision Avoidance of UAV Using Dueling Double Deep Q Network with Estimated Depth through CNN

11/2019 International Conference on Intelligent Robots and Systems (IROS), Macau, 2019 (poster-only option)

Target classification and prediction of unguided rocket trajectories using deep neural networks

● PROJECTS

03/2023 – CURRENT

Vision-Based Fast, Highly Reliable, and Agile Autonomous Drone Flight Using Deep Learning Approaches

Funded by National Research Fund (NRF)

06/2020 – CURRENT

Advanced Vehicle Autonomy & Intelligence Technology Development for Unmanned Vehicles

Funded by Korea Aerospace Research Institute (KARI)

12/2023 – 06/2024

Smart Crane Object Detection System

Funded by HD Hyundai Heavy Industry

07/2021 – 10/2023

State Estimation of a Moving Vehicle and Test Environment Development for Vision-Based Autonomous Landing of the UAV

Funded by Agency for Defense Development (ADD)

04/2020 – 11/2020

Virtual Workload Acceleration for Verification of Heterogeneous HW-Based Safety-Critical SW Platform

Funded by Electronics and Telecommunications Research (ETRI)

06/2019 – 09/2019

Object Detection Module from Pointcloud Data Using Machine Learning

Funded by Korea Institute of Machine & Materials (KIMM)

06/2018 – 05/2020

Research on Conflict-Free Methods between Guided Missiles Against Multiple Low-Altitude Rocket Targets

Funded by Agency for Defense Development (ADD)